

Yernar Mars

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Education

Nazarbayev Intellectual School of Astana (NISA)

IB Diploma Programme

Sep 2021 – May 2027

Astana, Kazakhstan

- HL: Mathematics (Analysis & Approaches), Digital Society, English B. SL: Physics, Computer Science, Russian A.
- “Orken” Presidential Merit Scholarship (government): full 6-year funding (~\$25,000); ranked 4th of 2,000+ nationally (top 0.2%).
- SAT: [SAT score] | IELTS: [IELTS score]

Honors & Awards

International & government-issued (Ministry of Education / DARYN / international brands)

- **Infomatrix Asia** (Ministry of Education), International — Bronze 2026 (6th of 3,000, **top 0.2%**, 13 countries) & 2025 (12th of 30, *KokysGet*).
- **World Robot Olympiad (WRO) 2026** — 2nd place (Astana); qualified for the National stage.
- **FIRST Tech Challenge (FTC) 2026** — Motivate, Reach, and Winning Alliance Captain awards; advanced to the Central Asia stage (21st of 40).
- **National Fibonacci Robot Olympiad** (Ministry of Education RK) — 3rd of 63 (2026); Judges’ Choice & 4th of 37 (2025), qualified for international finals in Antalya, Turkey.
- **National RSPC “Daryn”** (Ministry of Education) — 3rd place, Robotics & Intelligent Machines track (*SoilyGet*).
- **National “Bilim Shyny”** (DARYN, government), 2026 — Silver, 2nd of 300, “Technology of the Future.”

Startup & hackathon wins — 610,000+ KZT in cash prizes

- **1st place** — BECO National Hackathon 2025 (1/30; 150,000 KZT); Synergy Hackathon 2026 (100,000 KZT); Young Visionaries Shark Tank 2025 (1/21; 60,000 KZT); BrightFuture National Hackathon 2024 (1/20; 50,000 KZT).
- **3rd place** — Caspian Startup Battle 2026 (AOO NIS; 3/300; 100,000 KZT); International Startup Cup 2026 (3/320, 4 countries); National NIS Hack 2026 (AOO NIS; 3/110).
- **2nd place** — IT FEST Startup Battle 2025 (2/20; 100,000 KZT); Oylab Startup Battle (2/113; 50,000 KZT).

Selective programs (2026): Pioneer Academics Research · LaunchX Entrepreneurship · NYU Pre-College.

Research

Minimum-dose, contrast-free pulmonary-hypertension screening

Aug 2024 – Present

Lead student researcher — mentored by 7 Nazarbayev University professors (Kazakhstan’s #1 university) and a UMC Heart Center cardiologist

Astana, KZ

- Building an AI pipeline that measures pulmonary-artery and aorta diameters on **low-dose, non-contrast chest CT** and computes the PA:aorta ratio (a validated marker for pulmonary hypertension) — to find the minimum radiation dose at which screening stays reliable, avoiding both contrast dye and unnecessary radiation.
- Aorta segmentation held **12/12 integrity across a 2.6× dose range** (7.84–20.75 mGy) and 3 reconstruction kernels; ascending-aorta diameter measured to ± 1.7 –2.8 mm within-scan precision (~ 5 –7% CV) on a 3,673-file DICOM dataset.
- Built the PA:aorta ratio + PH-flag pipeline (PA model license-gated, pending). To be validated in radiologists’ labs toward clinical use; selected to present at the national *Zhas & Healthy* symposium.

Engineering Projects

SoilyGet — Autonomous soil-monitoring robot

Mar 2024 – Present

Lead hardware & systems engineer

Astana, KZ

- Built a **full-sized**, four-legged wheeled robot that autonomously augers soil samples to a **25 cm** depth, with a servo-driven **4-joint** manipulator modeled in **Fusion 360** and custom **3D-printed** parts. pXRF + field sensors feed an AI contamination map; first-authored the research paper and placed **3rd** at the National RSPC “Daryn” competition (Robotics & Intelligent Machines).

KokysGet — Vision-based waste sorter

Jan 2024 – Present

Founder — hardware, ML & deployment

Astana, KZ

- Trained a custom **YOLOv8** object-detection model on a self-labeled waste dataset (~ 800 annotated images) to classify recyclables in real time.

- Deployed end-to-end: an **ESP32-C3** with a Raspberry Pi Camera Module streams frames to a **Raspberry Pi** running inference; servo actuators sort each item into the correct bin; custom **3D-printed (STL)** chassis.
- Won National BECO Hackathon 2025; Bronze, Infomatrix Asia 2025.

Leadership & Experience

Mentoria — Peer-to-peer STEM academy & national hackathon

Co-founder & Director

Dec 2025 – Present
Astana, KZ (*Remote*)

- Founded a peer-to-peer STEM academy; grew an opportunities channel to 1,195 subscribers and peer-teaching groups (294 + 40 members) across math, physics, chemistry, biology, programming, SAT, and IELTS.
- Organized a free national student hackathon — 330 teams registered — hosted at Nazarbayev University, with a 1,000,000 KZT prize pool and a Bahandi developer internship for the winning team, funded by Bahandi (a major Kazakhstani fast-food chain).

AI automation (freelance)

Founder & developer

2023 – Present
Remote

- Build and sell LLM-based reply and booking assistants for clinics (website chat + Instagram/Facebook DMs); revenue helps cover family expenses and self-funds robotics prototypes.

QApps — EdTech startup: centralized university-admissions platform (\$2M valuation)

Software engineering intern

Summer 2026
Astana, KZ

- Selected as 1 of 3 interns from 30 nationwide applicants; build and debug features in Next.js/Nest.js, test in Docker, and ship via pull requests merged by a mentor. Contributed during growth to 4,000 users and \$50K ARR.

FIRST Tech Challenge — Team Spirit

Lead programmer (Java)

Oct 2025 – May 2026
Astana, KZ

- Programmed the competition robot (Java); **5–0** league record, **5–3** at the Central Asia stage (21st of 40).

Shenber — Summer STEM Camp

Co-founder & instructor

Summer 2026
Astana, KZ

- Co-founded *Shenber*, a paid summer STEM camp: taught 33 children (60,000 KZT each; ~2,000,000 KZT revenue), ~5 hours/day; led curriculum and instruction.

Robotics instructor & Jasyl El community service

Teacher & volunteer

2023 – 2026
Astana, KZ

- Teach robotics to children ages 8–14 — 20 students across 3 cohorts (80,000 KZT/month); completed 250 hours of paid environmental field work with Jasyl El (2 months, ~30 hrs/week, 60,000 KZT/month).

Skills

Programming: Python, Java, C/C++ (Arduino), JavaScript (Next.js / Nest.js). **ML / AI:** PyTorch, YOLOv8, nnU-Net / TotalSegmentator, CNNs, computer vision, LLM application building. **Tools:** Docker, Git, Flask, ESP32 / Raspberry Pi, 3D printing, Onshape & Fusion 360 (15+ CAD models). **Languages:** Kazakh (native), Russian (fluent), English (fluent, IB English B HL).